# Brutal Series B24 - Physical & Chemical Changes / Comparing Quantities / Respiration / Exponents & Powers

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

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**1.** When iron rusts, 56 parts of iron combine with 24 parts of oxygen by mass to form rust. If 90 grams of rust is collected, how much iron had rusted, and is rusting reversible? [Physical & Chemical Changes]

- **(A)** 34 grams of iron; not reversible
- **(B)** 63 grams of iron; not reversible
- **(C)** 80 grams of iron; not reversible
- **(D)** 90 grams of iron; not reversible
- **(E)** 27 grams of iron; not reversible
- **(F)** 63 grams of iron; reversible

**2.** A jacket is marked at 2000 rupees. The shop first takes off 20 percent, and then a further 10 percent off the reduced price. What is the final price paid? [Comparing Quantities]

- **(A)** 1380 rupees
- **(B)** 1440 rupees
- **(C)** 1400 rupees
- **(D)** 1600 rupees
- **(E)** 1500 rupees
- **(F)** 1480 rupees

**3.** An adult breathes about 18 times each minute at rest. During hard exercise the rate rises to 30 times each minute. How many more breaths are taken in 5 minutes of exercise than in 5 minutes of rest? [Respiration in Organisms]

- **(A)** 150 more breaths
- **(B)** 90 more breaths
- **(C)** 60 more breaths
- **(D)** 240 more breaths
- **(E)** 30 more breaths
- **(F)** 12 more breaths

**4.** Simplify and give the value: 2^3 x 2^4 / 2^5. [Exponents & Powers]

- **(A)** 32
- **(B)** 12
- **(C)** 8
- **(D)** 16
- **(E)** 4
- **(F)** 64
- **(G)** 2

**5.** Magnesium burns in air, combining with oxygen in the mass ratio 3:2 to give white magnesium oxide. To make 60 grams of magnesium oxide, what mass of magnesium is burnt? [Physical & Chemical Changes]

- **(A)** 50 grams
- **(B)** 20 grams
- **(C)** 30 grams
- **(D)** 40 grams
- **(E)** 36 grams
- **(F)** 45 grams
- **(G)** 24 grams

**6.** A trader buys a cycle for 800 rupees and sells it at a 20 percent profit. The buyer later sells it again at a 10 percent loss. What is the final selling price of the cycle? [Comparing Quantities]

- **(A)** 936 rupees
- **(B)** 864 rupees
- **(C)** 848 rupees
- **(D)** 960 rupees
- **(E)** 880 rupees
- **(F)** 720 rupees
- **(G)** 800 rupees

**7.** After sprinting hard, an athlete's leg muscles ache. This happens because the muscles, short of oxygen, break down glucose by anaerobic respiration to form which substance? [Respiration in Organisms]

- **(A)** Oxygen and lactic acid
- **(B)** Starch
- **(C)** Only carbon dioxide and water
- **(D)** Carbon dioxide and alcohol
- **(E)** Lactic acid
- **(F)** Alcohol and carbon dioxide
- **(G)** Glucose and oxygen

**8.** Work out the value of the expression (3^2)^3 divided by 3^4, using the index laws. [Exponents & Powers]

- **(A)** 9
- **(B)** 6
- **(C)** 3
- **(D)** 1
- **(E)** 27
- **(F)** 81
- **(G)** 243

**9.** A 250 gram block of ice is melted to water, and then the water is boiled away to steam. Counting both steps, how does the total mass change and how many of the steps are physical changes? [Physical & Chemical Changes]

- **(A)** Mass becomes zero; both physical
- **(B)** Mass stays 250 grams; both steps are physical
- **(C)** Mass stays 250 grams; one physical one chemical
- **(D)** Mass rises above 250 grams; both physical
- **(E)** Mass falls below 250 grams; one physical
- **(F)** Mass rises above 250 grams; both chemical
- **(G)** Mass stays 250 grams; both chemical

**10.** In a school of 360 students, 60 percent take part in sports, and of those, one quarter play cricket. How many students play cricket? [Comparing Quantities]

- **(A)** 90
- **(B)** 27
- **(C)** 60
- **(D)** 48
- **(E)** 54
- **(F)** 135
- **(G)** 216

**11.** While making bread, yeast is mixed into the dough and it rises. The gas that puffs up the dough, and the process that makes it, are: [Respiration in Organisms]

- **(A)** Carbon dioxide, made by anaerobic respiration (fermentation)
- **(B)** Lactic acid gas, made by fermentation
- **(C)** Oxygen, made by aerobic respiration
- **(D)** Carbon dioxide, made by breathing through lungs
- **(E)** Carbon dioxide, made by photosynthesis
- **(F)** Water vapour, made by aerobic respiration
- **(G)** Hydrogen, made by anaerobic respiration

**12.** By writing each factor as a power of 3, rewrite the product 81 times 27 as one single power of 3. [Exponents & Powers]

- **(A)** 3^11
- **(B)** 3^10
- **(C)** 9^7
- **(D)** 3^7 + 1
- **(E)** 3^7
- **(F)** 3^12
- **(G)** 3^4

**13.** On heating, 100 grams of marble chips release 44 grams of carbon dioxide gas. How much carbon dioxide comes from 25 grams of chips, and what kind of change is this breakdown? [Physical & Chemical Changes]

- **(A)** 22 grams; a chemical change
- **(B)** 11 grams; a chemical change
- **(C)** 9 grams; a chemical change
- **(D)** 4.4 grams; a chemical change
- **(E)** 25 grams; a chemical change
- **(F)** 44 grams; a chemical change
- **(G)** 11 grams; a physical change

**14.** The number of boys to girls in a class is in the ratio 3:5, and there are 320 students in all. If 40 more boys join, what is the new ratio of boys to girls? [Comparing Quantities]

- **(A)** 3:5
- **(B)** 4:5
- **(C)** 1:1
- **(D)** 2:3
- **(E)** 4:3
- **(F)** 7:8
- **(G)** 5:4

**15.** When a person breathes out into limewater, the limewater slowly turns milky. What does this show about the air we breathe out? [Respiration in Organisms]

- **(A)** Exhaled air is the same as inhaled air
- **(B)** Exhaled air is richer in carbon dioxide than inhaled air
- **(C)** Exhaled air contains lime
- **(D)** Exhaled air is richer in oxygen than inhaled air
- **(E)** Exhaled air is rich in hydrogen
- **(F)** Exhaled air has no gases at all
- **(G)** Exhaled air is rich in nitrogen only

**16.** Write the number 47000 in standard form. [Exponents & Powers]

- **(A)** 4.7 x 10^4
- **(B)** 4.700 x 10^4
- **(C)** 4.7 x 10^3
- **(D)** 47 x 10^4
- **(E)** 4.7 x 10^5
- **(F)** 47 x 10^3
- **(G)** 0.47 x 10^5

**17.** Water expands by about 9 percent of its volume when it freezes. If 200 millilitres of water freezes completely, what is the approximate volume of ice, and is freezing physical or chemical? [Physical & Chemical Changes]

- **(A)** About 290 millilitres; physical
- **(B)** About 209 millilitres; physical
- **(C)** About 182 millilitres; physical
- **(D)** About 218 millilitres; physical
- **(E)** About 218 millilitres; chemical
- **(F)** About 218 millilitres; reversible chemical
- **(G)** About 200 millilitres; physical

**18.** After a discount of 15 percent, a school bag sells for 1700 rupees. What was its marked price before the discount? [Comparing Quantities]

- **(A)** 2000 rupees
- **(B)** 1700 rupees
- **(C)** 1955 rupees
- **(D)** 1445 rupees
- **(E)** 1955.50 rupees
- **(F)** 2125 rupees
- **(G)** 1850 rupees

**19.** Each breath moves about 500 millilitres of air in and the same out. At 20 breaths per minute, how many litres of air are breathed in over one minute? [Respiration in Organisms]

- **(A)** 10 litres
- **(B)** 10000 litres
- **(C)** 100 litres
- **(D)** 1 litre
- **(E)** 20 litres
- **(F)** 0.5 litres

**20.** Find the value of (-2)^3 x (-2)^2. [Exponents & Powers]

- **(A)** -64
- **(B)** 64
- **(C)** -32
- **(D)** -16
- **(E)** -8
- **(F)** 32
- **(G)** 16

**21.** A box of 40 matchsticks is taken; 25 of them are struck and the chemical on the tips burns. What fraction of the matchsticks have undergone a chemical change, and is that change reversible? [Physical & Chemical Changes]

- **(A)** 2/5 of them; not reversible
- **(B)** 5/8 of them; reversible
- **(C)** 3/8 of them; not reversible
- **(D)** 25/40 of them; reversible
- **(E)** 15/40 of them; not reversible
- **(F)** 5/8 of them; not reversible
- **(G)** 5/8 of them; physical change

**22.** A sum of 8000 rupees is lent at 12.5 percent per year simple interest for 2 years. What is the total amount to be paid back? [Comparing Quantities]

- **(A)** 10000 rupees
- **(B)** 10800 rupees
- **(C)** 12000 rupees
- **(D)** 11000 rupees
- **(E)** 2000 rupees
- **(F)** 9600 rupees
- **(G)** 9000 rupees

**23.** Which word equation correctly shows what happens during respiration in the cells of living things? [Respiration in Organisms]

- **(A)** Glucose + nitrogen gives carbon dioxide + energy
- **(B)** Glucose gives oxygen + energy only
- **(C)** Water + energy gives glucose + oxygen
- **(D)** Oxygen + water gives glucose + energy
- **(E)** Glucose + carbon dioxide gives oxygen + energy
- **(F)** Carbon dioxide + water gives glucose + oxygen
- **(G)** Glucose + oxygen gives carbon dioxide + water + energy

**24.** Find the value of 2^10 / 2^7 + 3^0. [Exponents & Powers]

- **(A)** 1
- **(B)** 16
- **(C)** 10
- **(D)** 8
- **(E)** 11
- **(F)** 9
- **(G)** 2

**25.** Four events are listed: (i) curd forming from milk, (ii) ice melting, (iii) a cut apple turning brown, (iv) salt dissolving in water. How many of these are chemical changes? [Physical & Chemical Changes]

- **(A)** All except melting
- **(B)** Two and a half
- **(C)** Zero
- **(D)** Four
- **(E)** One
- **(F)** Two

**26.** A shopkeeper marks goods 25 percent above cost price and then allows a 20 percent discount on the marked price. What is the overall profit or loss percent? [Comparing Quantities]

- **(A)** 20 percent profit
- **(B)** 45 percent profit
- **(C)** 5 percent loss
- **(D)** 2.5 percent profit
- **(E)** 5 percent profit
- **(F)** 25 percent loss
- **(G)** No profit, no loss

**27.** A fish kept in a bowl keeps opening and closing its mouth and gill covers. What is the fish actually doing? [Respiration in Organisms]

- **(A)** Taking in carbon dioxide to make food
- **(B)** Taking in nitrogen dissolved in water
- **(C)** Using its mouth as a lung filled with air
- **(D)** Passing water over its gills to take in oxygen dissolved in the water
- **(E)** Pushing out lactic acid through its gills
- **(F)** Drinking water to make oxygen inside its body
- **(G)** Breathing air directly from above the water

**28.** Which is greater, 2^5 or 5^2, and by how much? [Exponents & Powers]

- **(A)** 2^5 is greater, by 7
- **(B)** They are equal
- **(C)** 5^2 is greater, by 25
- **(D)** 5^2 is greater, by 7
- **(E)** 5^2 is greater, by 3
- **(F)** 2^5 is greater, by 32

**29.** Photosynthesis in a leaf uses carbon dioxide and water to make glucose and release oxygen. Pick the statement that is fully correct about this process. [Physical & Chemical Changes]

- **(A)** It is a chemical change because new substances (glucose and oxygen) are formed
- **(B)** It is a chemical change in which no new substance forms
- **(C)** It is a physical change because the leaf stays green
- **(D)** It is a physical change that releases oxygen
- **(E)** It is a chemical change that can be easily reversed by cooling
- **(F)** It is neither physical nor chemical, just a mixing
- **(G)** It is a physical change because only water moves about

**30.** A town's population is 25000. In one year it rises by 10 percent, and the next year it falls by 10 percent. What is the population after the two years? [Comparing Quantities]

**(A)** 25250
**(B)** 24250
**(C)** 27500
**(D)** 24500
**(E)** 25000
**(F)** 24750
**(G)** 22500

**31.** A person sleeps for 8 hours breathing 15 times a minute, and stays awake for 16 hours breathing 20 times a minute. About how many breaths are taken in this whole day? [Respiration in Organisms]

**(A)** 12600 breaths
**(B)** 33600 breaths
**(C)** 21600 breaths
**(D)** 26400 breaths
**(E)** 19200 breaths
**(F)** 25200 breaths

**32.** Express the product 2^3 x 3^3 as a single power. [Exponents & Powers]

**(A)** 6^1
**(B)** 2^3
**(C)** 6^3
**(D)** 6^9
**(E)** 6^6
**(F)** 36^3
**(G)** 5^3

**33.** An iron nail weighing 50 grams is coated with a thin layer of zinc weighing 4 grams to stop it rusting. What is the coated nail's mass, and what is this rust-proofing method called? [Physical & Chemical Changes]

**(A)** 58 grams; galvanisation
**(B)** 54 grams; evaporation
**(C)** 54 grams; rusting
**(D)** 54 grams; crystallisation
**(E)** 54 grams; galvanisation
**(F)** 50 grams; painting
**(G)** 46 grams; galvanisation

**34.** Which of these four values is the largest: 3/5, 62 percent, 0.59, or 5/8? [Comparing Quantities]

**(A)** 62 percent
**(B)** 3/5
**(C)** 3/5 and 0.59 tie
**(D)** They are all equal
**(E)** 5/8
**(F)** 0.59
**(G)** 62 percent and 5/8 tie

**35.** Which one of these breathes through tiny pores called spiracles that lead into air tubes inside the body? [Respiration in Organisms]

- **(A)** A flowering plant
- **(B)** An insect such as a cockroach
- **(C)** A fish
- **(D)** An earthworm
- **(E)** A whale
- **(F)** A human

**36.** Find the value of (5^0 + 4^0)^2. [Exponents & Powers]

- **(A)** 1
- **(B)** 0
- **(C)** 9
- **(D)** 4
- **(E)** 8
- **(F)** 16
- **(G)** 2

**37.** A burning candle shows two changes at once: solid wax near the flame melts to liquid, and wax at the wick burns away. Which describes these two changes correctly? [Physical & Chemical Changes]

- **(A)** Both are physical
- **(B)** Melting is physical; burning is also physical
- **(C)** Both are reversible chemical changes
- **(D)** Neither is a change
- **(E)** Melting is physical; burning is chemical
- **(F)** Both are chemical
- **(G)** Melting is chemical; burning is physical

**38.** A man buys 12 pens for 144 rupees and sells each pen for 15 rupees. What is his profit percent? [Comparing Quantities]

- **(A)** 25 percent
- **(B)** 12 percent
- **(C)** 36 percent
- **(D)** 20 percent
- **(E)** 15 percent
- **(F)** 30 percent
- **(G)** 3 percent

**39.** An earthworm has no lungs or gills, yet it respires. How does it take in oxygen, and why must its surroundings stay damp? [Respiration in Organisms]

- **(A)** Through its moist skin; oxygen dissolves in the moisture before passing in
- **(B)** Through gills; the soil water flows over them
- **(C)** Through leaves on its skin; they need water
- **(D)** Through spiracles; dust must be kept out
- **(E)** Through its tail only; it must stay wet
- **(F)** Through tiny lungs; they need damp air
- **(G)** Through its mouth; it swallows air bubbles

**40.** A sheet of paper is folded so its thickness doubles with every fold; after n folds it is 2^n times the original. A sheet 0.1 millimetres thick is folded 8 times. What is its final thickness? [Exponents & Powers]

- **(A)** 1.6 millimetres
- **(B)** 12.8 millimetres
- **(C)** 0.8 millimetres
- **(D)** 25.6 millimetres
- **(E)** 51.2 millimetres
- **(F)** 256 millimetres

**41.** Hot water at 90 degrees Celsius is left to cool in a room. It cools by 5 degrees Celsius every 3 minutes. After 18 minutes, what is its temperature, and is cooling a physical or chemical change? [Physical & Chemical Changes]

- **(A)** 60 degrees Celsius; reversible chemical
- **(B)** 72 degrees Celsius; physical
- **(C)** 55 degrees Celsius; physical
- **(D)** 60 degrees Celsius; physical
- **(E)** 30 degrees Celsius; physical
- **(F)** 65 degrees Celsius; physical
- **(G)** 60 degrees Celsius; chemical

**42.** A second-hand cycle is bought for 2400 rupees, then 600 rupees is spent on repairs. It is finally sold for 3600 rupees. What is the profit percent on the total cost? [Comparing Quantities]

- **(A)** 40 percent
- **(B)** 30 percent
- **(C)** 20 percent
- **(D)** 25 percent
- **(E)** 12 percent
- **(F)** 16 percent
- **(G)** 50 percent

**43.** Some students claim 'plants do not respire, they only make food.' Why is this claim wrong? [Respiration in Organisms]

- **(A)** Plants take in carbon dioxide during respiration
- **(B)** Plants store food so they need no respiration
- **(C)** Plants respire only in their flowers
- **(D)** Plants only respire at night and never make food
- **(E)** Plants respire all the time, taking in oxygen and giving out carbon dioxide, like all living things
- **(F)** Plants breathe out oxygen during respiration
- **(G)** Plants make food but never need any energy

**44.** Solve for the whole number x: 2^x = 64. [Exponents & Powers]

- **(A)** 6
- **(B)** 5
- **(C)** 4
- **(D)** 8
- **(E)** 16
- **(F)** 32
- **(G)** 7

**45.** A solution is made by stirring blue copper sulphate into water; later the water is slowly evaporated and shiny blue crystals appear. Name the two physical changes used here, in order. [Physical & Chemical Changes]

- **(A)** Dissolving, then combustion
- **(B)** Crystallisation, then dissolving
- **(C)** Melting, then freezing
- **(D)** Dissolving, then a chemical reaction
- **(E)** Burning, then crystallisation
- **(F)** Dissolving, then crystallisation

**46.** At what rate of simple interest per year will 5000 rupees grow to 6200 rupees in 3 years? [Comparing Quantities]

- **(A)** 8 percent
- **(B)** 12 percent
- **(C)** 9 percent
- **(D)** 6 percent
- **(E)** 24 percent
- **(F)** 10 percent
- **(G)** 4 percent

**47.** During inhalation (breathing in), what happens to the diaphragm and the ribs, and what does this do to the chest space? [Respiration in Organisms]

- **(A)** The diaphragm flattens and the ribs move up and out, so the chest space increases
- **(B)** The diaphragm domes up and the ribs move in, so the chest space increases
- **(C)** The diaphragm domes up and the ribs move out, so the chest space decreases
- **(D)** The diaphragm and ribs do not move during breathing in
- **(E)** The diaphragm flattens and the ribs move in, so the chest space decreases
- **(F)** The diaphragm flattens and the ribs move up, so the chest space decreases

**48.** Light travels about 3 x 10^8 metres every second. About how far does it travel in 100 seconds, written in standard form? [Exponents & Powers]

- **(A)** 3 x 10^6 metres
- **(B)** 3 x 10^10 metres
- **(C)** 30 x 10^9 metres
- **(D)** 3 x 10^9 metres
- **(E)** 3 x 10^16 metres
- **(F)** 300 x 10^8 metres
- **(G)** 3 x 10^800 metres

**49.** When dilute acid is poured on baking soda, bubbles of a gas rush out and put out a burning splint. Identify the gas and say whether the change is physical or chemical. [Physical & Chemical Changes]

- **(A)** Carbon dioxide; chemical change
- **(B)** Nitrogen; chemical change
- **(C)** Carbon dioxide; physical change
- **(D)** Water vapour; physical change
- **(E)** Carbon dioxide; reversible physical change
- **(F)** Oxygen; chemical change
- **(G)** Hydrogen; chemical change

**50.** A 2 litre bottle of fruit drink is 15 percent sugar by volume. How many millilitres of sugar does the bottle contain? [Comparing Quantities]

- **(A)** 1700 millilitres
- **(B)** 30 millilitres
- **(C)** 3000 millilitres
- **(D)** 200 millilitres
- **(E)** 300 millilitres
- **(F)** 250 millilitres
- **(G)** 150 millilitres

**51.** Yeast can respire two ways. With plenty of oxygen it respires aerobically; with little oxygen it ferments. Which row correctly compares the energy released? [Respiration in Organisms]

- **(A)** Aerobic respiration releases no energy at all
- **(B)** Aerobic respiration releases slightly less energy
- **(C)** Fermentation releases more energy than aerobic respiration
- **(D)** Fermentation releases energy but aerobic respiration releases none
- **(E)** Both release exactly the same energy
- **(F)** Aerobic respiration releases much more energy than fermentation from the same glucose
- **(G)** Both release no energy, only gas

**52.** Which is the largest among 10^3, 3^5, and 2^9? [Exponents & Powers]

- **(A)** 10^3 and 2^9 tie
- **(B)** They are all equal
- **(C)** 3^5
- **(D)** 3^5 is largest at 343
- **(E)** 10^3
- **(F)** 2^9 and 3^5 tie

**53.** A 30 gram strip of pure wax is heated. 18 grams of it burns at the wick while 12 grams only melts and drips into a tray as solid wax again. What fraction of the original wax underwent a chemical change? [Physical & Chemical Changes]

- **(A)** 1/2
- **(B)** 18/100
- **(C)** 2/5
- **(D)** 3/10
- **(E)** 12/30
- **(F)** 2/3
- **(G)** 3/5

**54.** Pack A holds 500 grams for 90 rupees; Pack B holds 750 grams for 126 rupees. Which pack is cheaper per kilogram, and by how much? [Comparing Quantities]

- **(A)** Pack B is cheaper, by 36 rupees per kilogram
- **(B)** They cost the same per kilogram
- **(C)** Pack A is cheaper, by 6 rupees per kilogram
- **(D)** Pack A is cheaper, by 18 rupees per kilogram
- **(E)** Pack A is cheaper, by 12 rupees per kilogram
- **(F)** Pack B is cheaper, by 12 rupees per kilogram
- **(G)** Pack B is cheaper, by 24 rupees per kilogram

**55.** What is the key difference between breathing and respiration? [Respiration in Organisms]

**(A)** Respiration happens only in the lungs; breathing in every cell

**(B)** Breathing and respiration mean exactly the same thing

**(C)** Breathing is the physical taking in and giving out of air; respiration is the release of energy from food inside cells

**(D)** Breathing happens only in plants; respiration only in animals

**(E)** Respiration is the movement of air; breathing makes energy

**(F)** Breathing releases energy; respiration only moves air

**(G)** Breathing makes food; respiration makes oxygen

**56.** Find the value of (-1)^100 + (-1)^101. [Exponents & Powers]

**(A)** 201

**(B)** 0

**(C)** -1

**(D)** 1

**(E)** 100

**(F)** 2

**(G)** -2

**57.** Which single list contains ONLY chemical changes? [Physical & Chemical Changes]

**(A)** Melting of wax, burning of paper, rusting of iron

**(B)** Burning of paper, rusting of iron, cooking of food

**(C)** Souring of milk, melting of ice, rusting of iron

**(D)** Burning of paper, rusting of iron, cutting of paper

**(E)** Cooking of food, dissolving of sugar, burning of paper

**(F)** Freezing of water, rusting of iron, burning of paper

**(G)** Boiling of water, souring of milk, cooking of food

**58.** A student scores 480 marks out of 600. The pass mark is 35 percent. By how many marks did the student clear the pass mark? [Comparing Quantities]

**(A)** 300 marks

**(B)** 210 marks

**(C)** 165 marks

**(D)** 330 marks

**(E)** 270 marks

**(F)** 480 marks

**(G)** 80 marks

**59.** During exercise, both the breathing rate and the heartbeat rate go up together. This happens mainly to: [Respiration in Organisms]

**(A)** Supply more carbon dioxide to the muscles

**(B)** Cool the body down by losing water only

**(C)** Make more food in the muscles

**(D)** Stop the muscles from respiring

**(E)** Supply more oxygen to the muscles and remove more carbon dioxide from them

**(F)** Add nitrogen to the blood faster

**60.** Simplify and give the value of (2^2)^3 x 2 / 2^4. [Exponents & Powers]

- **(A)** 16
- **(B)** 64
- **(C)** 2
- **(D)** 8
- **(E)** 4
- **(F)** 32
- **(G)** 6

**61.** Curd is set from milk by adding a spoon of old curd, and the milk turns sour over a few hours. Choose the fully correct description. [Physical & Chemical Changes]

- **(A)** A chemical change that can be reversed by heating
- **(B)** A chemical change that is easily reversed by cooling
- **(C)** A change in which no new substance forms
- **(D)** A physical change that gives out a gas
- **(E)** A slow chemical change that cannot be reversed
- **(F)** A physical change because milk is still a liquid
- **(G)** A fast physical change that can be reversed

**62.** Shop A offers a flat 40 percent off. Shop B offers 25 percent off and then a further 20 percent off. On an item priced 1000 rupees, which shop's final price is lower? [Comparing Quantities]

- **(A)** They are equal at 600 rupees
- **(B)** Shop B is lower at 600; Shop A is 700
- **(C)** Shop A at 600; Shop B at 550
- **(D)** Shop A is lower at 550
- **(E)** Shop B is lower at 550
- **(F)** Shop A is lower at 600; Shop B is 700
- **(G)** They are equal at 650 rupees

**63.** A leaf takes in and gives out gases through tiny openings on its surface. What are these openings called, and what gas does the leaf take in during respiration? [Respiration in Organisms]

- **(A)** Stomata; carbon dioxide
- **(B)** Gills; oxygen
- **(C)** Pores called tracheae; oxygen
- **(D)** Stomata; nitrogen
- **(E)** Lenticels only; nitrogen
- **(F)** Spiracles; oxygen
- **(G)** Stomata; oxygen

**64.** Simplify and give the value of 3^4 x 3^2 / 3^5. [Exponents & Powers]

- **(A)** 27
- **(B)** 1
- **(C)** 6
- **(D)** 3
- **(E)** 3^11
- **(F)** 9
- **(G)** 81

**65.** A shopkeeper finds that an iron sheet of mass 800 grams gains 5 percent in mass after rusting in the rainy season. What is the mass of rust formed on it? [Physical & Chemical Changes]

- **(A)** 80 grams
- **(B)** 840 grams
- **(C)** 760 grams
- **(D)** 4 grams
- **(E)** 40 grams
- **(F)** 45 grams

**66.** A quantity A is 25 percent more than another quantity B. By what percent is B less than A? [Comparing Quantities]

- **(A)** 30 percent
- **(B)** 20 percent
- **(C)** 25 percent
- **(D)** 12.5 percent
- **(E)** 15 percent
- **(F)** 40 percent
- **(G)** 75 percent

**67.** At rest a man breathes 18 times a minute. During exercise this value doubles. How many breaths does he take during 10 minutes of exercise, and how many more is that than 10 minutes at rest? [Respiration in Organisms]

- **(A)** 360 breaths; 360 more than at rest
- **(B)** 180 breaths; 90 more than at rest
- **(C)** 360 breaths; 90 more than at rest
- **(D)** 36 breaths; 18 more than at rest
- **(E)** 180 breaths; 180 more than at rest
- **(F)** 360 breaths; 180 more than at rest
- **(G)** 720 breaths; 360 more than at rest

**68.** Find the value of (7^3 x 7^0) / 7^2. [Exponents & Powers]

- **(A)** 14
- **(B)** 0
- **(C)** 7
- **(D)** 1
- **(E)** 7^5
- **(F)** 49
- **(G)** 343

**69.** Which of the following is the BEST evidence that a chemical change has taken place rather than a physical one? [Physical & Chemical Changes]

- **(A)** The material is cut into smaller pieces
- **(B)** A new substance with new properties is formed
- **(C)** The material is squeezed into a smaller space
- **(D)** The material is dissolved in water
- **(E)** The material changes its shape
- **(F)** The material only changes its temperature

**70.** In how many years will 6000 rupees earn 1440 rupees as simple interest at 6 percent per year? [Comparing Quantities]

- **(A)** 8 years
- **(B)** 4 years
- **(C)** 4.5 years
- **(D)** 2 years
- **(E)** 6 years
- **(F)** 3 years

**71.** Why does holding your breath for a while create a strong urge to breathe again? [Respiration in Organisms]

- **(A)** Lactic acid leaves the body too fast
- **(B)** Oxygen builds up and forces you to breathe
- **(C)** The lungs fill completely with fresh air
- **(D)** Carbon dioxide builds up in the body and triggers the urge to breathe it out
- **(E)** Water vapour builds up in the lungs
- **(F)** The heart stops beating for a while
- **(G)** Nitrogen levels fall too low in the blood

**72.** Write the standard-form number 6.02 x 10^5 as an ordinary number. [Exponents & Powers]

- **(A)** 60200000
- **(B)** 602000
- **(C)** 60200
- **(D)** 620000
- **(E)** 602000.0 with a different value
- **(F)** 6.02000

**73.** Crystals of a salt are grown from a hot saturated solution as it cools slowly overnight, giving 36 grams of crystals from a beaker. If three identical beakers are set up, how many grams of crystals form in total, and is crystallisation physical or chemical? [Physical & Chemical Changes]

- **(A)** 12 grams; physical
- **(B)** 108 grams; chemical
- **(C)** 108 grams; reversible chemical
- **(D)** 120 grams; physical
- **(E)** 72 grams; physical
- **(F)** 108 grams; physical
- **(G)** 36 grams; physical

**74.** A phone's price is 12000 rupees before tax. An 18 percent GST is added. What is the final price including tax? [Comparing Quantities]

- **(A)** 13800 rupees
- **(B)** 14160 rupees
- **(C)** 12018 rupees
- **(D)** 15600 rupees
- **(E)** 2160 rupees
- **(F)** 14000 rupees
- **(G)** 12180 rupees

**75.** Gills in fish and air tubes in insects are built differently but share an important feature that makes them good at their job. What is it? [Respiration in Organisms]

- **(A)** A large surface area for fast exchange of gases
- **(B)** A bony shell to protect the heart
- **(C)** A waterproof layer to stop oxygen entering
- **(D)** A green colour for making food
- **(E)** A store of food for the winter
- **(F)** A small surface area to save energy
- **(G)** A hard, dry covering to keep gases out

**76.** Find the value of 2^4 + 2^3 + 2^0. [Exponents & Powers]

- **(A)** 17
- **(B)** 24
- **(C)** 9
- **(D)** 25
- **(E)** 16
- **(F)** 11
- **(G)** 27

**77.** A green leaf and a candle flame are compared. Which pair of changes is correctly matched as chemical? [Physical & Chemical Changes]

- **(A)** Water moving up the leaf (chemical) and wax burning (chemical)
- **(B)** Water moving up the leaf (chemical) and wax melting (chemical)
- **(C)** Food made in the leaf (chemical) and wax burning at the wick (chemical)
- **(D)** Leaf bending in wind (chemical) and wax burning (chemical)
- **(E)** Food made in the leaf (physical) and wax burning (physical)
- **(F)** Food made in the leaf (chemical) and wax melting (chemical)

**78.** A sum of 5600 rupees is shared between A and B in the ratio 3:5. How much more does B receive than A? [Comparing Quantities]

- **(A)** 1400 rupees
- **(B)** 1750 rupees
- **(C)** 1120 rupees
- **(D)** 3500 rupees
- **(E)** 2800 rupees
- **(F)** 700 rupees

**79.** In an experiment, soaked germinating seeds in a sealed flask are found to use up oxygen and give out a gas that turns limewater milky, while the flask gets slightly warm. What does this show? [Respiration in Organisms]

- **(A)** The seeds are burning their stored fat with a flame
- **(B)** The seeds are giving out oxygen as they respire
- **(C)** The seeds are respiring, using oxygen and giving out carbon dioxide and heat
- **(D)** The seeds are not alive and nothing is happening
- **(E)** The seeds are only absorbing water, nothing else
- **(F)** The seeds are doing photosynthesis in the dark

**80.** Which standard-form number is larger: 3.2 x 10^6 or 9.9 x 10^5? [Exponents & Powers]

- **(A)** Both equal 3200000
- **(B)** 3.2 x 10^6 is larger
- **(C)** 9.9 x 10^5 is larger by 10 times
- **(D)** 9.9 x 10^5 is larger because 9.9 > 3.2
- **(E)** They are equal
- **(F)** 9.9 x 10^5 is larger
- **(G)** Cannot be compared

**81.** In a school demo, 5 grams of a fuel burns completely and gives 15 grams of carbon dioxide plus 9 grams of water vapour, taking in oxygen from the air. What mass of oxygen was used, by conservation of mass? [Physical & Chemical Changes]

- **(A)** 10 grams
- **(B)** 20 grams
- **(C)** 19 grams
- **(D)** 6 grams
- **(E)** 24 grams
- **(F)** 4 grams
- **(G)** 14 grams

**82.** Two ratios are compared: 4:5 and 7:9. Which is greater? [Comparing Quantities]

- **(A)** 4:5 is greater
- **(B)** 7:9 is greater by 1
- **(C)** 7:9 is greater
- **(D)** Cannot be compared
- **(E)** They are equal
- **(F)** 4:5 is smaller
- **(G)** 4:5 = 7:9 = 0.8

**83.** Roots of a plant also need to respire. Why do most plants die if their soil is waterlogged for a long time? [Respiration in Organisms]

- **(A)** Water fills the air spaces in the soil, so the roots cannot get oxygen to respire
- **(B)** Water washes away all the carbon dioxide the roots need
- **(C)** The roots make too much food and burst
- **(D)** Roots do not respire, so water cannot harm them
- **(E)** Water gives the roots extra oxygen, which poisons them
- **(F)** Water makes the roots breathe out lactic acid

**84.** Simplify and give the value of (5^3 x 5^4) / (5^2 x 5^2). [Exponents & Powers]

- **(A)** 1
- **(B)** 5
- **(C)** 625
- **(D)** 5^3 x 2
- **(E)** 125
- **(F)** 3125
- **(G)** 25

**85.** A rusty gate is rubbed clean and then painted. Which statement about preventing rust is correct? [Physical & Chemical Changes]

- **(A)** Paint chemically turns rust back into iron
- **(B)** Paint adds oxygen, which speeds rusting
- **(C)** Paint makes the iron rust faster but look better
- **(D)** Paint cools the iron so it cannot rust
- **(E)** Paint keeps air and moisture away, so it slows rusting
- **(F)** Paint reverses the chemical change of rusting
- **(G)** Rusting needs only oxygen, so paint does nothing

**86.** An article is sold for 990 rupees at a profit of 10 percent. What was its cost price? [Comparing Quantities]

- **(A)** 1000 rupees
- **(B)** 990 rupees
- **(C)** 909 rupees
- **(D)** 900 rupees
- **(E)** 891 rupees
- **(F)** 1089 rupees

**87.** Place these in the correct path that air follows when a person breathes in: nostrils, windpipe, lungs, nasal cavity. [Respiration in Organisms]

- **(A)** Nostrils, windpipe, nasal cavity, lungs
- **(B)** Lungs, windpipe, nasal cavity, nostrils
- **(C)** Nostrils, lungs, windpipe, nasal cavity
- **(D)** Nasal cavity, windpipe, nostrils, lungs
- **(E)** Nasal cavity, nostrils, lungs, windpipe
- **(F)** Nostrils, nasal cavity, windpipe, lungs

**88.** Express the product (2 x 3)^3 / 3^3 as a single power, and give its value. [Exponents & Powers]

- **(A)** 2^3, which is 8
- **(B)** 3^3, which is 27
- **(C)** 2^9, which is 512
- **(D)** 6^3, which is 216
- **(E)** 6^0, which is 1
- **(F)** 2^6, which is 64
- **(G)** 2^0, which is 1

**89.** Equal masses of wood are treated two ways: one piece is sawn into sawdust, the other is burnt to ash. Which row correctly classifies the two and compares the change in substance? [Physical & Chemical Changes]

- **(A)** Both are chemical because the wood is destroyed
- **(B)** Sawing is chemical; burning is physical
- **(C)** Burning is reversible; sawing is irreversible
- **(D)** Sawing is physical; burning is also physical
- **(E)** Sawing is physical (same wood); burning is chemical (new ash and gases)
- **(F)** Both make the same new substance
- **(G)** Both are physical because only the size changes

**90.** A price rises from 80 rupees to 100 rupees. Later it falls back from 100 rupees to 80 rupees. What are the percent rise and the percent fall? [Comparing Quantities]

- **(A)** Rise of 25 percent, then a fall of 24 percent
- **(B)** Rise of 25 percent, then a fall of 20 percent
- **(C)** Rise of 20 percent, then a fall of 20 percent
- **(D)** Rise of 25 percent, then a fall of 16 percent
- **(E)** Rise of 25 percent, then a fall of 25 percent
- **(F)** Rise of 20 percent, then a fall of 25 percent

**91.** A frog can breathe in two ways: through its lungs and through its moist skin. While it stays underwater in a pond for a long time, which way of breathing does it depend on most? [Respiration in Organisms]

- **(A)** Through its moist skin, taking in dissolved oxygen
- **(B)** Through its lungs, taking in air from the water
- **(C)** It stops respiring completely underwater
- **(D)** Through spiracles like an insect
- **(E)** Through gills like a fish
- **(F)** Through its lungs, which fill with pond water
- **(G)** Through its eyes, which absorb oxygen

**92.** Express 1000000 (one million) as a power of 10, and state how many zeros it has. [Exponents & Powers]

- **(A)** 10^7, with 7 zeros
- **(B)** 10^6, with 6 zeros
- **(C)** 6^10, with 6 zeros
- **(D)** 10^3, with 3 zeros
- **(E)** 10^6, with 5 zeros
- **(F)** 100^3, with 6 zeros
- **(G)** 10^5, with 5 zeros

**93.** A student lists signs seen during a change: a colour change, a gas given off, heat released, and a new smell. How many of these four are typical signs of a chemical change? [Physical & Chemical Changes]

- **(A)** One
- **(B)** Two
- **(C)** Four
- **(D)** Zero
- **(E)** Three
- **(F)** Only the gas
- **(G)** Only heat and gas

**94.** A discount of 12 percent is offered on a marked price. A customer pays 1320 rupees after the discount. How much money was saved by the discount? [Comparing Quantities]

- **(A)** 180 rupees
- **(B)** 1500 rupees
- **(C)** 132 rupees
- **(D)** 160 rupees
- **(E)** 158.40 rupees
- **(F)** 240 rupees
- **(G)** 158 rupees

**95.** Compared with the air we breathe in, the air we breathe out has: [Respiration in Organisms]

**(A)** More oxygen, more carbon dioxide, and less water vapour
**(B)** Less oxygen, more carbon dioxide, and more water vapour
**(C)** More oxygen, less carbon dioxide, and less water vapour
**(D)** No oxygen at all and only carbon dioxide
**(E)** The same amounts of every gas
**(F)** Less carbon dioxide and more oxygen
**(G)** Less oxygen, less carbon dioxide, and no water vapour

**96.** Find the value of (a^2)^3 x a^0 / a^5 when a = 2. [Exponents & Powers]

**(A)** 16
**(B)** 0
**(C)** 2
**(D)** 1
**(E)** 32
**(F)** 4
**(G)** 8

**97.** A 60 gram piece of copper is heated strongly in air and turns into black copper oxide weighing 75 grams. What mass of oxygen has combined with the copper, and what kind of change is this? [Physical & Chemical Changes]

**(A)** 15 grams; a physical change
**(B)** 135 grams; a chemical change
**(C)** 60 grams; a chemical change
**(D)** 15 grams; a chemical change
**(E)** 5 grams; a chemical change
**(F)** 45 grams; a chemical change
**(G)** 75 grams; a chemical change

**98.** A man sells two watches for 1200 rupees each. On one he makes a 20 percent profit and on the other a 20 percent loss. Taking both together, what is his overall profit or loss? [Comparing Quantities]

**(A)** A loss of 4 rupees only
**(B)** A profit of 50 rupees
**(C)** A profit of 100 rupees
**(D)** A loss of 200 rupees
**(E)** No profit, no loss
**(F)** A loss of 100 rupees
**(G)** A loss of 50 rupees

**99.** A nurse counts a resting patient's breaths for just 15 seconds to save time and gets a count of 4. What is the breathing rate per minute, and is it normal for an adult at rest? [Respiration in Organisms]

**(A)** 16 breaths per minute; no, far too fast
**(B)** 8 breaths per minute; yes, normal
**(C)** 60 breaths per minute; no, too fast
**(D)** 16 breaths per minute; no, too slow
**(E)** 4 breaths per minute; no, too slow
**(F)** 16 breaths per minute; yes, normal for an adult at rest
**(G)** 20 breaths per minute; yes, normal

**100.** Find the value of $4^3 / 2^4$ by first writing 4 as a power of 2. [Exponents & Powers]

**(A)** 16

**(B)** 1

**(C)** 2

**(D)** 8

**(E)** 4

**(F)** 64

**(G)** 32